

COLLEGE OF COMPUTER AND INFORMATION SCIENCE

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IT Practicum

Duane Jancis O. Bacayon

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Overview of the Company

Ecolab is a global leader in water, hygiene and infection prevention solutions and services that protect people and the resources vital to life. For more than a century, Ecolab has advanced innovation by integrating science-based solutions, data-driven insights, AI technology and world-class service. This unique combination enables Ecolab to partner with customers to define their operations, helping them achieve peak performance.

Vision

Our vision is to be the global leader in water, hygiene and antimicrobial technologies and services - providing and protecting what is vital: clean water, safe food and healthy environments.

Mission

Our mission is to be the leading global innovator, developer, and provider of cleaning, sanitation and maintenance products, systems, and services. As a team, we will achieve aggressive growth and fair return for our shareholders. We will accomplish this by exceeding the expectations of our customers while conserving resources and preserving the quality of the environment.

Company Background

Ecolab Inc. is a global company specializing in water, hygiene, and infection prevention solutions and services. It helps businesses maintain clean and safe environments, improve operations, protect public health, and manage resources such as water and energy. Today, Ecolab serves customers in approximately 170 countries.

- 1920s - In 1923, Merritt J. Osborn founded Economics Laboratory (EL) in St. Paul, Minnesota. The company initially focused on cleaning solutions for hotels. In 1924, it entered the warewashing industry with Soilax, a product designed for mechanical dishwashing. By 1928, the company introduced its first detergent dispensing system, combining cleaning products with equipment to improve efficiency and reduce waste.

- 1960s - During the 1960s, Ecolab continued to expand its cleaning and sanitation services while developing new products and technologies for businesses. The company also strengthened its international presence and expanded its services to support a wider range of industries and customers.

- 1970s - During the 1970s, Ecolab expanded its industrial and institutional cleaning solutions. It developed specialized products and services for industries such as food service, hospitality, healthcare, and manufacturing, helping customers improve cleanliness, safety, and operational efficiency.

- 1980s - During the 1980s, Ecolab continued its growth through innovation, acquisitions, and market expansion. The company broadened its cleaning, sanitation, and pest-management services while increasing its presence across different industries and international markets.

- 2000s - During the 2000s, Ecolab increased its focus on water conservation, sustainability, and advanced technology. The company introduced solutions such as 3D TRASAR™ Technology, which helped customers monitor and manage water use, improve efficiency, and reduce environmental impact.

- 2020s - During the 2020s, Ecolab strengthened its focus on water, hygiene, infection prevention, and sustainability. During the COVID-19 pandemic, the company provided cleaning and infection-prevention solutions to help organizations maintain safer environments. Ecolab also continued advancing its 2030 Positive Impact goals to protect people, conserve resources, and reduce environmental impact.

source: https://www.ecolab.com/en-us/about/our-history

Discussion of the Nature of Tasks

The nature of the tasks performed during the internship primarily involved IT operations, technical support, asset management, and end-user assistance under the Digital Client Service Department of Ecolab Philippines, Inc. At the beginning of the internship, the tasks focused on orientation and familiarization with the company's IT processes, including the Digital Help Center and ServiceNow used for handling employee incidents and requests. As the internship progressed, the responsibilities became more hands-on, particularly in preparing, resetting, configuring, and deploying laptops for newly hired and existing employees. These tasks included reinstalling operating systems using bootable USB drives, creating and configuring user accounts, installing necessary applications and updates, diagnosing hardware components, and ensuring that devices were ready for deployment. The work therefore required careful attention to procedures and accuracy because the devices and services being handled were directly used by employees.

Another major aspect of the internship involved troubleshooting, inventory management, and IT administrative support. Various technical issues were encountered, such as laptops experiencing performance problems, microphone and power issues, Windows update failures, application installation problems, and difficulties connecting to the company's private Wi-Fi. Troubleshooting activities involved checking hardware through diagnostic tools, updating Windows, clearing temporary files, configuring system and network settings, using command-line commands, and coordinating unresolved concerns with the appropriate networking or technical support team. The internship also included maintaining accurate records of IT assets by checking physical laptops against Excel inventory lists, verifying serial numbers, recording device conditions, and updating asset information through the company's service website. In addition, I assisted with procurement-related activities by requesting vendor quotations, preparing approval forms, communicating with managers and requestors, and sending purchase orders to vendors. These responsibilities demonstrated that IT operations involve not only technical problem-solving but also documentation, coordination, and proper management of organizational resources.

The tasks also provided opportunities for continuous learning and professional development through both practical experience and formal training. During work-from-home days, I completed online courses through Coursera and Cisco Networking Academy, including topics related to technical support, computer networking, hardware and update support, and digital safety and security awareness. These learning activities complemented the hands-on tasks performed in the workplace and helped strengthen my understanding of IT support and cybersecurity-related practices. The internship also required communication with employees, supervisors, managers, vendors, and other support teams through platforms such as Microsoft Teams and Outlook, allowing me to develop professional communication and coordination skills. Overall, the nature of the internship tasks was diverse and practical, combining technical support, troubleshooting, asset management, procurement, documentation, and continuous learning. These experiences allowed me to apply concepts learned from my Information Technology studies in an actual organizational environment while developing greater patience, adaptability, problem-solving skills, and responsibility in handling IT-related tasks.

Software Used

1. **Zscaler** - A cloud-based security platform that protects users and devices by monitoring and securing internet and network access.
2. **MS Teams** - A communication and collaboration application used for messaging, meetings, file sharing, and coordination.
3. **MS Outlook** - An email and communication application used for sending emails, managing schedules, and coordinating with employees and vendors.
4. **MS Azure** - Microsoft's cloud computing platform that provides services for identity, applications, storage, networking, and other IT operations.
5. **MS OneNote** - A digital note-taking application used to organize notes, information, and documentation.
6. **MS Word** - A word-processing application used to create, edit, and format documents.
7. **MS Excel** - A spreadsheet application used to organize, track, analyze, and manage data such as IT asset inventories.
8. **MS OneDrive** - A cloud storage service used to store, synchronize, and share files across devices.
9. **MS Edge** - A web browser used to access websites, web-based company systems, and online services.
10. **Windows Autopilot** - A Microsoft service used to simplify and automate the setup and deployment of Windows devices.
11. **ServiceNow** - An IT service management platform used to create, monitor, track, and manage service requests and incidents.
12. **Command Prompt** - A Windows command-line interface used to execute commands for system administration and troubleshooting.
13. **Nexthink** - An IT monitoring and digital employee experience platform used to monitor the health and performance of devices and users.
14. **Bitlocker** - A Windows security feature that encrypts storage drives to protect data from unauthorized access.

15. **Cyberark** - A cybersecurity platform used to securely manage and protect privileged accounts, credentials, and access.
16. **Company Portal** - A company-managed application used to access organizational resources, applications, device settings, and management services.
17. **Wireshark** - A network protocol analyzer used to capture and examine network traffic for troubleshooting and analysis.
18. **UEFI** - A modern firmware interface that initializes computer hardware and provides the environment for starting the operating system.
19. **PrinterLogic** - A printer management platform used to manage and deploy printers and their configurations to users and devices.
20. **Windows Creation Tool** - A Microsoft utility used to create Windows installation media and assist with Windows installation or reinstallation.
21. **Password Vault** - A secure system used to store, manage, and protect passwords and other authentication credentials.
22. **Nalco Password Generation Tool** - A company-specific tool used to generate or manage passwords for authorized IT-related purposes.
23. **Cisco Unified Communications Manager Administration** - An administrative platform used to manage and configure Cisco communication systems, including IP phones and related services.

Hardware Used

1. **MS Teams Rooms** - A meeting-room system that integrates a camera, microphone, speaker, touch screen, and Mini PC for virtual meetings.
2. **Company-owned User Devices** - IT-managed devices such as laptops, monitors, and mobile phones provided to employees for work purposes.
3. **Peripherals** - External computer accessories such as mice and keyboards used for user interaction and productivity.
4. **Switches** - Network devices that connect multiple devices within a network and forward data to the appropriate destination.
5. **Routers** - Network devices that connect different networks and direct data traffic between them.
6. **IP Phones** - Phones that use an IP network to transmit voice communication instead of traditional telephone lines.
7. **Multi-Function Printer** - A device that combines printing with functions such as scanning, copying, and sometimes faxing.
8. **Barcode Printer** - A specialized printer used to produce barcode labels for identifying and tracking assets or products.
9. **Barcode Scanner** - A device used to scan barcodes and quickly capture the information encoded in them.
10. **RF Scanner** - A wireless scanning device used to capture and process information from assets or items through radio-frequency connectivity.
11. **Smart Televisions** - Internet-connected display devices used for presentations, meetings, and multimedia purposes.
12. **LED Display** - An electronic display that uses light-emitting diodes to present visual information, presentations, or announcements.

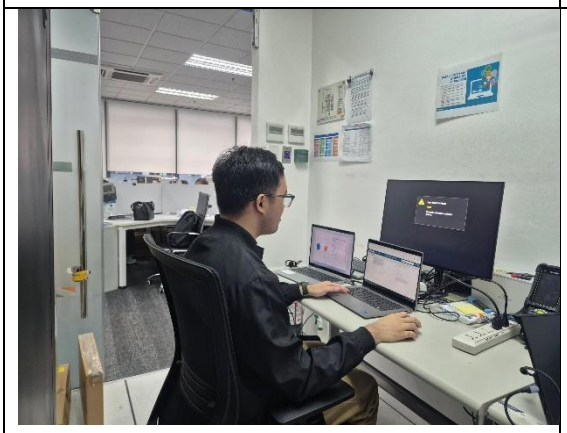
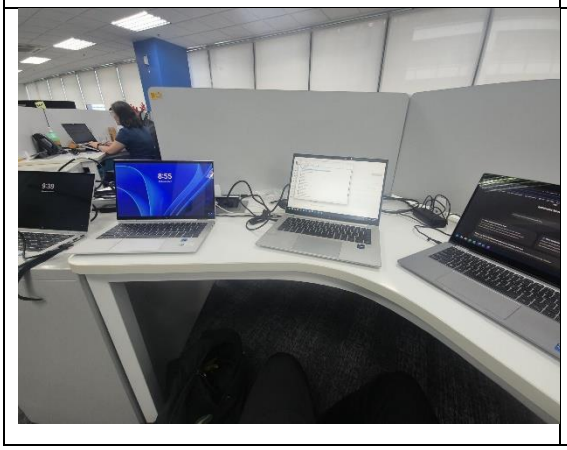
Practicum Outputs

The practicum resulted in several outputs related to IT operations and technical support. These included prepared and configured laptops for new employees and replacement users, reset and reinstalled operating systems, installed necessary applications and Windows updates, and performed hardware diagnostics to ensure that devices were functional before deployment. The practicum also involved updating asset information for returned, deployed, and stored devices and maintaining inventory records using Microsoft Excel.

Another significant output was the completion of various IT-related administrative tasks. These included processing purchase requests for peripherals, requesting quotations from vendors, preparing approval documents, communicating with managers and the finance department, and sending purchase orders to vendors. Accountability forms for employees were also collected, monitored, and submitted to the supervisor.

The practicum also produced solutions to various technical issues encountered by employees. These included troubleshooting laptop performance, application installation, microphone problems, Windows update issues, and private Wi-Fi connectivity. For issues that could not be resolved through initial troubleshooting, tickets were escalated to the appropriate networking service. In addition, the practicum included the completion of online technical courses covering computer networking, hardware and update support, and digital safety and security awareness, resulting in certificates and digital badges.

Actual Photos

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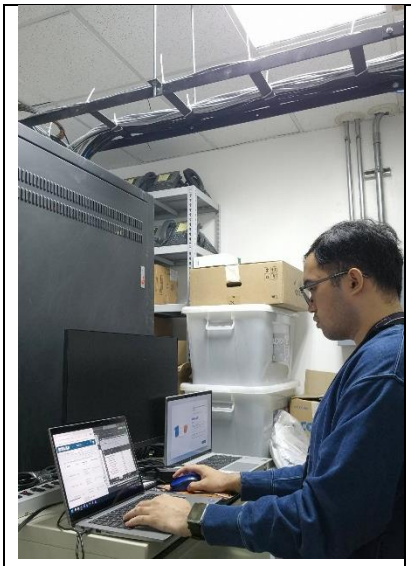
1. & 9.) On 25th of August, I participated with Digital-Client Services Team of Ecolab Philippines for the plantation visit located in Biñan City and Calamba City. Our purpose for this visit is to configure the multi-functional printers and barcode printers by resetting an OS and installing drivers. We make sure that these devices are working, as well as the internet connection.

2. & 4.) On 23rd of July, I attended the UAS Unitech 2026 in Pasig City as an Ecolab Philippines representative, together with my supervisor. The event creates interaction between different tech companies focusing on the trends and new innovations specifically for Artificial Intelligence. The event consists of seminars, booths, catering, and social interactions.

3.) One of my main tasks is procurement for the IT goods and services such as the orders of the laptops, monitors, peripherals, and cables. It was a long task in order to obtain an item with different steps such as quotation request to the vendor, purchase approval to the finance department, and item approval to the requestor's manager.

5.) Most of the time, I deployed several laptops for the new employees, requestors, and their temporary use around different Ecolab sites in the Philippines. The main purpose is to reset the PCs using the bootable USB flash drive and configure different settings for more efficiency. At the same time, I troubleshoot some of the technical issues encountered by employees.

6.) The main conference area is held during the general meetings, training, monthly report, and party. I was one of the team members who set up and controlled the devices, including smart televisions, cameras, microphones, and speakers. We make sure that the ongoing presentation is smooth and deliverable, especially during the MS Teams call and PowerPoint presentation.

		
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7.) During the one-week long problem of private Wi-Fi connectivity in a whole office, I was assigned to troubleshoot each of some employee's laptops in order to update the internet security platform, check the company compliance, and other performances.

8.) I used multi-purpose printers several times, mostly for printing (such as accountability form and IT-related documents) and scanning (such as delivery receipt and photocopy). The scanned documents as a soft copy can be directed through employee's email.

10.) Most of the previously used and new laptops are stored in a stockroom with different areas depending on the laptop's condition, such as break-fix laptops, reserved laptops, old laptops, and brand-new laptops. Each of those laptops have attached note paper that contains the name of the last user and the return date.

11.) As the supervisor provided the excel list of the laptop needed to dispose, I double checked the brand, model, and serial number each of those physical units as part of the yearly laptop disposal. I took a note about the missing physical units and missing list of laptops. The IT asset disposal provider will collect the disposed laptops as their right to destruct safely.

12.) As part of the procurement process, when I received the order of multiple brand-new monitors, I notified the financial representative and my supervisor about the new arrival of items and send necessary files such as delivery receipt.

Synthesis of the Practicum Engagement

The practicum engagement provided comprehensive experience in the field of Information Technology by exposing the student to actual IT operations and end-user support. Throughout the 62-day engagement, I participated in laptop deployment, operating system installation, account configuration, application installation, hardware diagnostics, troubleshooting, inventory management, procurement assistance, and technical documentation. These activities demonstrated how IT personnel contribute to maintaining reliable and efficient technology services within an organization.

One of the most significant aspects of the practicum was the opportunity to handle real technical problems. Issues involving device performance, hardware, software, Windows updates, authentication, and network connectivity required systematic troubleshooting and persistence. The experience also demonstrated the importance of knowing when to escalate an issue to a specialized team, particularly when dealing with network-related problems affecting multiple users.

Beyond technical skills, the practicum strengthened professional skills such as communication, documentation, organization, patience, problem-solving, and responsibility. Coordination with employees, supervisors, managers, vendors, finance personnel, and other IT associates was necessary to complete tasks successfully. The experience also emphasized the importance of maintaining accurate records and following established procedures when handling company devices and user information.

Overall, the practicum served as an important transition from academic learning to professional IT practice. It allowed me to apply classroom knowledge in a real organizational environment while gaining a broader understanding of IT support, asset management, networking, cybersecurity, and administrative operations. The experience developed both technical competence and workplace readiness and provided a stronger foundation for future work in the Information Technology and cybersecurity fields.